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كلية الحاسبات وتقنية المعلومات
جامعة الملك عبدالعزيز

CPIT 240- Project | **Phase #1**

(E-sports Platform Database System)

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Spring 2025

Project Overview:

The **E-sports Platform Database System** is designed to manage players, teams, tournaments, and match results in a university or online gaming environment. It stores essential information about participants, tracks scores and rankings, and supports tournament organization and performance analytics.

Description of the Structure and Requirements

Structure of the Database:

The system consists of several main entities:

1. **Player**
   - **Attributes:** Player_ID, Name, Email, Username, Rank, Team_ID
   - **Description:** Stores personal and account information about each registered player.

2. **Team**
   - **Attributes:** Team_ID, Team_Name, Captain_ID, Game_Title
   - **Description:** Represents each team, its name, and the game it competes in. A team can have multiple players.

3. **Tournament**
   - **Attributes:** Tournament_ID, Tournament_Name, Start_Date, End_Date, Game_Title, Organizer
   - **Description:** Stores details about each tournament hosted on the platform.

4. **Match**
    - **Attributes:** Match_ID, Tournament_ID, Team1_ID, Team2_ID, Winner_ID, Match_Date
    - **Description:** Contains match details, including participating teams and match results.

5. **Scoreboard**
    - **Attributes:** Score_ID, Player_ID, Match_ID, Score, Kills, Assists, Deaths
    - **Description:** Tracks performance metrics for each player in every match.

6. **Game** *(optional)*
    - **Attributes:** Game_Title, Genre, Platform
    - **Description:** Lists supported games (e.g., Valorant, League of Legends, FIFA).

Key Requirements:

- Each player must belong to **one team**.
- A team can participate in **multiple tournaments**.
- A match belongs to **one tournament** and includes **two teams**.
- Each match result updates the **scoreboard** and **player statistics**.
- The system allows organizers to **create tournaments**, **add teams**, **schedule matches**, and **track results**.

Description of Database Operations

Typical database operations include:

Insert Operations

- Add new players, teams, games, and tournaments.
- Record match results and player performance.

Update Operations

- Update team rosters, player ranks, or tournament schedules.

Delete Operations

- Remove inactive players or outdated tournaments.

Select (Query) Operations

- Retrieve player rankings.
- Display upcoming matches and tournaments.
- Show team performance summaries.
- Generate leaderboards by tournament or game.

Constraints

Primary Keys

- Player_ID, Team_ID, Tournament_ID, Match_ID, and Score_ID uniquely identify records in their respective tables.

Foreign Keys

- Team_ID in *Player* references *Team*.
- Tournament_ID in *Match* references *Tournament*.
- Player_ID and Match_ID in *Scoreboard* reference *Player* and *Match*.

Value Constraints

- Score, Kills, Assists, and Deaths must be non-negative.
- Start_Date must be earlier than End_Date in tournaments.

Uniqueness Constraints

- Email and Username in *Player* must be unique.
- Team_Name and Tournament_Name must be unique.